

AYAN KUMAR

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SUMMARY

Computer engineering student with hands-on experience building retrieval-augmented generation pipelines, vector-database-backed retrieval systems, and full-stack applications. Contributor to open-source projects across testing, feature development, and refactoring work. Comfortable working across the stack, from model integration and data pipelines to production-facing web and mobile interfaces.

EDUCATION

Punjabi University, Patiala

Bachelor of Technology, Computer Engineering

Jul 2024 - Jul 2028

EXPERIENCE

Research Intern, Institute Incubation and Innovation Council (I3C), IIT Jammu

IIT Jammu Summer School 2026

Jun 2026 - Present

- Completing coursework and applied research projects in artificial intelligence and machine learning as part of a selective summer research program.

Open Source Contributor, GirlScript Summer of Code (GSSoC) 2026

devpath-web, commitpulse, arnio - Remote

Apr 2026 - Present

- Merged 9+ pull requests across three production repositories, including refactors, new features, and test coverage, reviewed under standard open-source contribution guidelines.
- Wrote Vitest test suites covering accessibility, visual consistency, and edge-case handling in commitpulse, and implemented CSV null-value handling, quality-profile statistics, and a new schema field in arnio.

PROJECTS

Anonymous Location-Based Social Platform

React Native, Firebase/Supabase, OpenStreetMap API

- Built a location-based social platform for anonymous, geotagged posts using manual location selection instead of exact GPS tracking, with anonymous authentication and real-time data handling.
- Designed gamification mechanics, including achievement badges, exploration rewards, and engagement-based progression, to increase user retention.
- Planned a scalable backend architecture with privacy-preserving analytics and monetization features, including enhanced customization and unlimited media uploads.

Smart Local Knowledge Assistant

TypeScript, Next.js, Gemini API, Vectra (vector database), pdf-parse

- Built a retrieval-augmented generation (RAG) assistant that grounds responses in government source documents using a local vector database, returning source-cited answers instead of unverified model output.
- Built a document ingestion pipeline that parses, chunks, embeds, and indexes PDF and text uploads on submission, with filtering by state and district.
- Added model routing between Gemini Pro and Flash with fallback handling, reducing median response time to under 2 seconds.
Live: smart-assistant-taupe.vercel.app

LLMVC - Prompt Version Control CLI

TypeScript, Node.js, SQLite, LangChain

- Built a local-first command-line tool for git-style versioning of LLM prompts (save, diff, rollback), storing full change history in SQLite across 5 LLM providers.
- Built a word-level diff engine and an automated evaluation runner scoring prompt revisions with a separate LLM as judge; fixed authentication and endpoint-routing bugs across provider integrations.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, HTML, CSS

Frameworks & Tools: Django, Node.js, React.js, Next.js, React Native, TensorFlow, scikit-learn, Git, Vercel

Data & ML: RAG, vector databases (Qdrant, Vectra), LangChain, LLM integration, SQLite, MongoDB, Pandas, NumPy

ADDITIONAL

- Solved 400+ data structures and algorithms problems on LeetCode, covering dynamic programming, graphs, trees, and greedy algorithms.
- Contributor to Gitree, a PyPI-published Python CLI tool for generating LLM context from codebases.
- Completed Elements of AI, GitHub Copilot Integration, EA Software Engineering Virtual Experience, and Gen AI Academy 2.0.